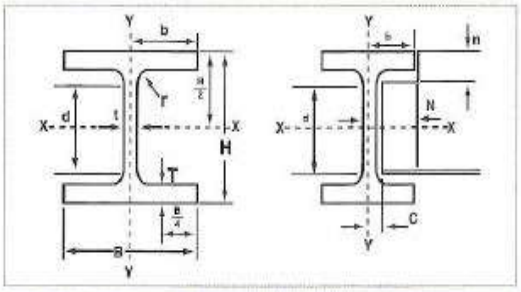
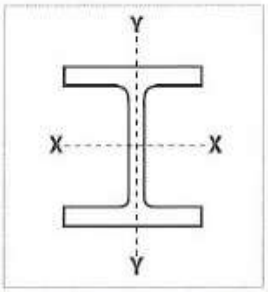
Dimensions and Properties



4a (i) Beams (Imperial Units) - Part I

	Designation Size		Unit Weight		Height Of Beam	Width Of Beam	Thickness		Root Radius	Depth Between Fillets	Area Of Section	Ratios For Local Buckling		Dimensions For Detailing		
							Flange	Web				Flange	Web	End Clearance	Notch	
					H	B	T	t	r	d	A	b/T	d/t	C	N	n
	in	(mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W14	14 x 16	(356 x 406)	342	509.0	445.5	415.5	62.7	39.1	15.2	289.6	649	3.31	7.40	22	198	78
			370	550.6	455.6	418.5	67.5	42.1	15.2	290.2	702	3.10	6.89	23	198	83
			398	592.3	464.6	421.4	72.3	45.0	15.2	289.7	755	2.92	6.44	24	198	87
			426	634.0	474.6	424.0	77.0	47.6	15.2	290.2	808	2.75	6.10	26	198	92
			455	677.1	483.1	427.6	81.5	51.2	15.2	289.6	863	2.62	5.66	28	198	97
			500	744.1	497.8	432.1	88.9	55.6	15.2	289.6	948	2.43	5.21	30	198	104
			550	818.5	514.1	436.9	97.0	60.5	15.2	289.6	1043	2.25	4.79	32	198	112
			605	900.3	531.4	442.3	105.6	65.9	15.2	289.8	1147	2.09	4.40	35	198	121
			665	989.6	549.7	448.3	114.8	71.9	15.2	289.7	1261	1.95	4.03	38	198	130
			730	1086.4	569.5	454.4	124.7	78.0	15.2	289.7	1385	1.82	3.72	41	198	140
W16	16 x 5 1/2	(406 x 140)	26	38.7	398.0	141.8	8.6	6.4	10.2	360.4	49.7	8.24	56.3	5	78	19
			31	46.1	403.2	142.2	11.2	6.8	10.2	360.4	58.6	6.35	53.0	5	78	21
	16 x 7	(406 x 178)	36	53.6	402.6	177.7	10.9	7.7	10.2	360.4	69.0	8.15	46.8	6	95	21
			40	59.5	406.4	177.9	12.8	7.9	10.2	360.4	76.5	6.95	45.6	6	95	23
			45	67.0	409.4	178.8	14.3	8.8	10.2	360.4	85.5	6.25	41.0	6	95	25
			50	74.4	412.8	179.5	16.0	9.5	10.2	360.4	94.5	5.61	37.9	7	95	26
			57	84.8	417.3	180.8	18.2	10.9	10.2	360.6	108	4.98	33.0	7	95	28
	16 x 10 1/4	(406 x 260)	67	99.7	414.8	260.0	16.9	10.0	10.2	360.6	127	7.70	36.0	7	135	27
			77	114.6	419.6	261.5	19.3	11.6	10.2	360.6	146	6.77	31.2	8	135	30
			89	132.4	425.4	263.3	22.2	13.3	10.2	360.6	169	5.92	27.0	9	135	32
			100	148.8	431.0	264.8	25.0	14.9	10.2	360.6	190	5.29	24.3	9	135	35
W18	18 x 6	(457 x 152)	35	52.1	449.8	152.4	10.9	7.6	10.2	407.6	66.6	6.99	53.6	6	82	21
			40	59.5	454.6	152.9	13.3	8.1	10.2	407.6	76.2	5.75	50.3	6	82	24
			45	67.0	458.0	153.8	15.0	9.0	10.2	407.6	85.6	5.13	45.3	7	82	25
			46	68.5	458.7	153.9	15.3	9.1	10.2	407.7	87.1	5.03	44.6	7	82	26
			50	74.4	462.0	154.4	17.0	9.6	10.2	407.6	94.5	4.54	42.5	7	82	27
			55	81.8	465.8	155.3	18.9	10.5	10.2	407.6	105	4.11	38.8	7	82	29
	18 x 7 1/2	(457 x 191)	41	61.0	450.0	189.0	10.8	8.1	10.2	408.4	76	8.75	50.4	6	750	21
			45	67.0	453.4	189.9	12.7	8.5	10.2	407.6	85.5	7.48	48.0	6	101	23
			50	74.4	457.0	190.4	14.5	9.0	10.2	407.6	94.6	6.57	45.3	7	101	25
			55	81.8	460.0	191.3	16.0	9.9	10.2	407.6	104	5.98	41.2	7	101	26
			60	89.3	463.4	191.9	17.7	10.5	10.2	407.6	114	5.42	38.8	7	101	28
			65	96.7	466.1	192.8	19.1	11.4	10.2	407.6	123	5.06	35.7	8	101	29
			66	98.2	467.2	192.8	19.6	11.4	10.2	407.6	125	4.92	35.8	8	101	30
			71	105.7	469.1	193.9	20.6	12.6	10.2	407.6	134	4.71	32.4	8	101	31
	18 x 11	(457 x 279)	76	113.1	462.5	280.3	17.3	10.8	10.2	407.6	144	8.12	37.7	7	145	27
			86	128.0	467.1	281.7	19.6	12.1	10.2	407.6	163	7.20	33.7	8	145	30
			97	144.4	472.2	283.1	22.1	13.6	10.2	407.6	184	6.40	30.0	9	145	32
			106	157.7	475.7	284.5	23.9	15.0	10.2	407.5	201	5.96	27.2	9	145	34
			119	177.1	481.8	286.1	26.9	16.6	10.2	407.6	226	5.31	24.5	10	145	37
			130	193.5	489.0	283.5	30.5	17.0	10.2	407.6	247	4.65	24.0	11	143	41
			143	212.8	495.0	285.0	33.5	18.5	10.2	407.5	271	4.25	22.0	11	143	44
			158	235.1	501.0	287.0	36.6	20.6	10.2	407.8	299	3.92	19.8	12	143	47
			175	260.4	509.0	289.0	40.4	22.6	10.2	408.2	331	3.58	18.1	13	143	50
W21	21 x 6 1/2	(533 x 165)	44	65.5	524.8	165.1	11.4	8.9	12.7	476.6	84	7.24	53.6	6	88	24
			50	74.4	529.1	165.9	13.5	9.7	12.7	476.7	95	6.14	49.4	7	88	26
			57	84.8	534.9	166.5	16.5	10.3	12.7	476.5	108	5.04	46.3	7	88	29

Note: Calculated based on 1 kg = 2.2046 lb and 1 m = 3.2808 feet.
*Please contact us for further clarification.



4a (i) Beams (Imperial Units) - Part II

Unit Weight		Surface Area per Metre	Second Moment Of Area		Radius Of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant	Designation Size	
			Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	u	x	H	J		
lb/ft	kg/m	m^2	cm^4	cm^4	cm	cm	cm^3	cm^3	cm^3	cm^3			dm^6	cm^4	in	
342	508.95	2.45	203800	75180	17.7	10.8	9151	3619	11010	5543	0.840	6.4	27.5	7401	**14 x 16**	**W14**
370	550.62	2.47	226900	82670	18.0	10.9	9962	3951	12080	6058	0.841	6.1	31.1	9240		
398	592.29	2.50	249800	90380	18.2	10.9	10750	4289	13130	6583	0.842	5.7	34.8	11350		
426	633.96	2.52	274800	98130	18.4	11.0	11580	4629	14240	7108	0.843	5.5	38.8	13720		
455	677.11	2.55	299400	106600	18.6	11.1	12400	4987	15340	7669	0.843	5.2	43.0	16460		
500	744.08	2.59	341900	120000	19.0	11.3	13740	5555	17160	8553	0.844	4.9	50.2	21390		
550	818.49	2.63	392400	135500	19.4	11.4	15260	6202	19260	9560	0.846	4.6	58.9	27870		
605	900.34	2.67	450600	153100	19.8	11.6	16960	6922	21610	10680	0.848	4.3	69.4	36130		
665	989.63	2.72	518100	173400	20.3	11.7	18850	7736	24250	11960	0.849	4.0	82.0	46760		
730	1086.36	2.77	597000	196300	20.8	11.9	20970	8640	27230	13370	0.851	3.8	97.1	60230		
26	38.69	1.33	12510	410	15.9	2.87	629	57.8	724	90.8	0.858	47.5	0.155	10.7	**16 x 5 1/2**	**W16**
31	46.13	1.34	15690	538	16.4	3.03	778	75.7	888	118	0.871	38.9	0.207	19.0		
36	53.57	1.48	18720	1021	16.5	3.85	930	115	1055	178	0.871	38.3	0.392	23.1	**16 x 7**	
40	59.53	1.49	21600	1203	16.8	3.97	1063	135	1199	209	0.880	33.8	0.466	33.3		
45	66.97	1.50	24330	1365	16.9	3.99	1189	153	1346	237	0.880	30.5	0.533	46.1		
50	74.41	1.51	27310	1546	17.0	4.04	1323	172	1501	267	0.882	27.6	0.608	62.8		
57	84.83	1.52	31520	1794	17.1	4.07	1511	198	1724	309	0.881	24.5	0.714	92.2		
67	99.71	1.83	39730	4951	17.7	6.25	1915	381	2128	581	0.888	25.4	1.96	99.4	**16 x 10 1/4**	
77	114.59	1.84	46110	5758	17.8	6.28	2198	440	2457	673	0.887	22.4	2.31	149		
89	132.45	1.86	54060	6768	17.9	6.33	2541	514	2859	788	0.887	19.7	2.75	227		
100	148.82	1.87	61830	7754	18.0	6.39	2869	586	3246	899	0.887	17.7	3.20	322		
35	52.09	1.48	21370	645	17.9	3.11	950	84.7	1096	133	0.859	43.9	0.311	21.4	**18 x 6**	**W18**
40	59.53	1.49	25500	795	18.3	3.23	1122	104	1287	163	0.868	37.5	0.387	33.8		
45	66.97	1.50	28930	913	18.4	3.27	1263	119	1453	187	0.869	33.6	0.448	47.7		
46	68.46	1.50	29530	933	18.4	3.27	1288	121	1482	191	0.869	33.0	0.458	50.4		
50	74.41	1.50	32670	1047	18.6	3.33	1414	136	1627	213	0.873	30.1	0.518	65.9		
55	81.85	1.51	36590	1185	18.7	3.37	1571	153	1811	240	0.873	27.4	0.591	89.2		
41	61.01	1.62	25380	1218	18.2	3.99	1128	129	1286	201	0.865	42.6	0.6	26	**18 x 7 1/2**	
45	66.97	1.63	29380	1452	18.5	4.12	1296	153	1471	237	0.872	37.9	0.705	37.1		
50	74.41	1.64	33320	1671	18.8	4.20	1458	176	1653	272	0.877	33.9	0.818	51.8		
55	81.85	1.65	37050	1871	18.8	4.23	1611	196	1831	304	0.877	30.9	0.922	69.2		
60	89.29	1.66	41020	2089	19.0	4.29	1770	218	2014	338	0.880	28.3	1.04	90.7		
65	96.73	1.66	44590	2281	19.0	4.30	1913	237	2184	369	0.879	26.3	1.14	114		
66	98.22	1.67	45730	2347	19.1	4.33	1957	243	2232	379	0.881	25.7	1.18	121		
71	105.66	1.67	48760	2507	19.0	4.32	2079	259	2383	404	0.877	24.4	1.26	145		
76	113.10	2.01	55460	6344	19.6	6.64	2398	453	2669	692	0.885	27.9	3.14	118	**18 x 11**	
86	127.98	2.02	63520	7295	19.7	6.69	2720	518	3039	792	0.886	24.8	3.65	170		
97	144.35	2.03	72710	8367	19.9	6.74	3079	591	3457	906	0.886	22.1	4.24	244		
106	157.75	2.04	79600	9178	19.9	6.76	3347	645	3775	991	0.884	20.5	4.68	311		
119	177.09	2.06	91040	10530	20.1	6.82	3779	736	4284	1132	0.885	18.4	5.44	442		
130	193.46	2.06	102500	11590	20.4	6.86	4192	818	4761	1257	0.891	16.6	6.09	603		
143	212.81	2.08	114400	12960	20.5	6.91	4624	910	5278	1400	0.891	15.2	6.90	800		
158	235.13	2.09	127300	14450	20.6	6.95	5083	1007	5839	1554	0.890	14.0	7.80	1059		
175	260.43	2.11	143700	16300	20.8	7.01	5646	1128	6525	1743	0.891	12.8	8.90	1423		
44	65.48	1.67	35030	859	20.5	3.20	1335	104	1561	166	0.847	47.0	0.566	31.9	**21 x 6 1/2**	**W21**
50	74.41	1.68	40810	1032	20.8	3.30	1543	124	1797	199	0.853	41.4	0.686	47.1		
57	84.83	1.69	48640	1276	21.2	3.44	1819	153	2107	243	0.862	35.5	0.857	73.8		

Note: Calculated based on 1 kg = 2.2046 lb and 1 m = 3.2808 feet.
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